OTEL-07: Dynatrace OTLP Integration

> **Series:** OTEL — OpenTelemetry Integration | **Notebook:** 7 of 8 | **Created:** January 2026 | **Last Updated:** 07/30/2026

Complete Setup for OpenTelemetry with Dynatrace

This notebook provides end-to-end configuration for sending OpenTelemetry data to Dynatrace, including authentication, endpoints, and verification.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS or Managed with OTLP enabled

Requirement	Details
Permissions	Token creation access
Knowledge	OTEL-01 through OTEL-06

Sprint 1.337 (April 2026): Major OpenTelemetry Updates

Sprint 1.337 brought several OpenTelemetry-relevant changes:

1. **OTel `service.name` enrichment** — Dynatrace now uses the OpenTelemetry `service.name` resource attribute to **enrich** the Dynatrace service entity rather than creating a parallel one. Smartscape displays `service.name (detected name)`; spans/metrics gain a `dt.service.name` field, queryable directly:

```
dql fetch spans, from:-1h | filter isNotNull(dt.service.name) | summarize span_count = count(), by:{dt.service.name, dt.smartscape.service}
```

This eliminates the historical "two services per process" pattern when OTel SDKs run alongside OneAgent.

2. **Ktor service technology recognized** — `KTOR_CLIENT` and `KTOR_SERVER` enum values added across request attributes, calculated metrics, and extension host availability. Ktor (Kotlin async HTTP framework) services now appear with explicit technology in Smartscape, no more `KOTLIN_GENERIC` fallback.

3. **OneAgent + OpenTelemetry-injector coexistence matrix** — newly documented in **K8S-11 § 2a** (added in Tier 2 Wave 0). Covers double-instrumentation symptoms, init-container ordering, `LD_PRELOAD` / `JAVA_TOOL_OPTIONS` clobber, trace-context conflicts, and a per-namespace decision flow. Reference this when planning OTel auto-instrumentation in environments where OneAgent is also deployed (ToDo #3).

4. **SDv2 unified rules for AWS Lambda** — Service Detection v2 now supports Lambda functions with unified rules for both OTel and OneAgent. New FaaS-specific metrics: invocation/failure counts, duration, trigger type breakdown. See **CLOUD-04 § Sprint 1.337**.

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1. Dynatrace OTLP Endpoints

SaaS Endpoints

Dynatrace supports **OTLP/HTTP only** for native ingest. gRPC is not supported for direct ingest — use a Collector to convert gRPC to HTTP.

Signal	HTTP Endpoint
All (base)	`https://{env-id}.live.dynatrace.com/api/v2/otlp`
Traces	`https://{env-id}.live.dynatrace.com/api/v2/otlp/v1/traces`
Metrics	`https://{env-id}.live.dynatrace.com/api/v2/otlp/v1/metrics`
Logs	`https://{env-id}.live.dynatrace.com/api/v2/otlp/v1/logs`

> **Important:** gRPC is **not supported** for direct Dynatrace ingest. If your SDKs use gRPC, route through a Collector with an `otlp` gRPC receiver and `otlphttp` exporter. See Transform OTLP gRPC.

ActiveGate Endpoints

For on-premises or network-restricted environments:

Signal	Endpoint
All	`https://{activegate-host}:9999/e/{env-id}/api/v2/otlp`

Dynatrace Collector Distribution

Dynatrace provides its own **Collector distribution** with verified, production-ready components:

Aspect	Detail
Image	`ghcr.io/dynatrace/dynatrace-otel-collector/dynatrace-otel-collector`
Features	Pre-configured for Dynatrace, upstream-compatible, monthly releases

Aspect	Detail
Docs	Dynatrace Collector

> **Tip:** The Dynatrace Collector distribution is recommended for production deployments. It includes verified components and stays current with upstream releases.

Environment ID

Find your environment ID:
1. Log into Dynatrace
2. Look at the URL: `https://abc12345.live.dynatrace.com`
3. `abc12345` is your environment ID

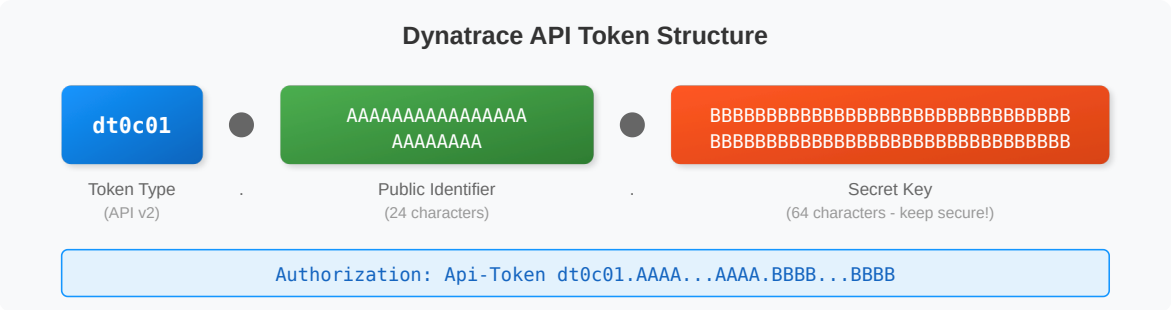
2. Authentication Setup

Create API Token

1. Navigate to **Settings > Access tokens**
2. Click **Generate new token**
3. Name: `otel-ingest-token`
4. Select scopes:

Scope	Signal	Required
`openTelemetryTrace.ingest`	Traces	Yes (for traces)
`metrics.ingest`	Metrics	Yes (for metrics)
`logs.ingest`	Logs	Yes (for logs)
`events.ingest`	Events	Optional

Token Format



Header Format

Use the `Api-Token` prefix in the Authorization header:

```
Authorization: Api-Token &amp;lt;your-full-token&amp;gt;
```

Token Type and Auth Scheme

The `Api-Token` scheme above is for **classic API Tokens** (prefix `dt0c01.*`). **Platform Tokens** (prefix `dt0u01.*` for user tokens, `dt0s16.*` for service-user tokens) use `Bearer ${TOKEN}` instead:

```
# Classic API Token
Authorization: Api-Token dt0c01.XXXXXXXX...

# Platform Token (Gen3-preferred)
Authorization: Bearer dt0u01.XXXXXXXX...
```

Using the wrong scheme returns **401 Unauthorized** even when the token has the correct scopes — a frequent first-deployment failure mode. Platform Tokens are preferred in Gen3 environments because they support fine-grained scopes via IAM policies. For OTLP ingest both token types work; the scope to grant is `metrics.ingest` (Platform Token) or the equivalent classic-token scope from the table above.

3. Collector Configuration

Image Versioning Best Practice

> **Important:** Always pin your OTel Collector image to a specific version. Using `latest` can cause unexpected behavior during upgrades.

```
# Avoid - Non-deterministic deployments
image: otel/opentelemetry-collector-contrib:latest

# Recommended - Pin to a specific version
image: otel/opentelemetry-collector-contrib:0.151.0

# Or use the Dynatrace distribution (pinned, not :latest)
image: ghcr.io/dynatrace/dynatrace-otel-collector/dynatrace-otel-
collector:v0.48.0
```

Check the OpenTelemetry Collector releases or Dynatrace Collector releases for the current stable version.

Complete Collector Config for Dynatrace

```
# otel-collector-config.yaml
receivers:
  otlp:
    protocols:
      grpc:
        endpoint: 0.0.0.0:4317
      http:
        endpoint: 0.0.0.0:4318

processors:
  batch:
    timeout: 10s
    send_batch_size: 1000

  memory_limiter:
    check_interval: 1s
    limit_mib: 800
    spike_limit_mib: 200

  # Add service.name if missing
  resource:
    attributes:
      - key: service.name
        value: "unknown-service"
        action: insert

exporters:
  otlphttp/dynatrace:
    endpoint: https://${DT_ENV_ID}.live.dynatrace.com/api/v2/otlp
    headers:
      Authorization: Api-Token ${DT_API_TOKEN}
    retry_on_failure:
      enabled: true
      initial_interval: 5s
      max_interval: 30s
      max_elapsed_time: 300s

  debug:
    verbosity: detailed

extensions:
  health_check:
    endpoint: 0.0.0.0:13133

service:
```

```
  extensions: [health_check]
  pipelines:
    traces:
      receivers: [otlp]
      processors: [memory_limiter, resource, batch]
      exporters: [otlphttp/dynatrace]
    metrics:
      receivers: [otlp]
      processors: [memory_limiter, resource, batch]
      exporters: [otlphttp/dynatrace]
    logs:
      receivers: [otlp]
      processors: [memory_limiter, resource, batch]
      exporters: [otlphttp/dynatrace]
```

Resource Sizing Guide

Environment	CPU Limit	Memory Limit	Batch Size
Dev/Test	200m	256Mi	500
Staging	500m	512Mi	1000
Production	1000m	1Gi	2000

Running with Environment Variables

```
export DT_ENV_ID="abc12345"
export DT_API_TOKEN="&lt;your-api-token&gt;"

otelcol-contrib --config otel-collector-config.yaml
```

4. Direct SDK Export

Python Direct to Dynatrace

```
import os
from opentelemetry import trace
from opentelemetry.sdk.trace import TracerProvider
from opentelemetry.sdk.trace.export import BatchSpanProcessor
from opentelemetry.exporter.otlp.proto.http.trace_exporter import
OTLPSpanExporter
from opentelemetry.sdk.resources import Resource

# Configuration
DT_ENV_ID = os.environ["DT_ENV_ID"]
DT_API_TOKEN = os.environ["DT_API_TOKEN"]

# Resource with service name
resource = Resource.create({
    "service.name": "my-python-app",
    "service.version": "1.0.0",
    "deployment.environment.name": "production"
})

# Setup tracer
provider = TracerProvider(resource=resource)
exporter = OTLPSpanExporter(
    endpoint=f"https://{DT_ENV_ID}.live.dynatrace.com/api/v2/otlp/v1/traces",
    headers={"Authorization": f"Api-Token {DT_API_TOKEN}"}
)
provider.add_span_processor(BatchSpanProcessor(exporter))
trace.set_tracer_provider(provider)
```

Java with System Properties

```
# Set environment variables first
export DT_ENV_ID="abc12345"
export DT_API_TOKEN="&amp;lt;your-api-token&amp;gt;"

# URL-encode the token for the header (space becomes %20)
java -javaagent:opentelemetry-javaagent.jar \
  -Dotel.service.name=my-java-app \
  -
Dotel.exporter.otlp.endpoint=https://${DT_ENV_ID}.live.dynatrace.com/api/v2/ot
lp \
  -Dotel.exporter.otlp.headers="Authorization=Api-Token%20${DT_API_TOKEN}" \
  -jar app.jar
```

Node.js

```
const { NodeSDK } = require('@opentelemetry/sdk-node');
const { OTLPTraceExporter } = require('@opentelemetry/exporter-trace-otlp-
http');
const { Resource } = require('@opentelemetry/resources');

const sdk = new NodeSDK({
  resource: new Resource({
    'service.name': 'my-node-app',
  }),
  traceExporter: new OTLPTraceExporter({
    url:
`https://${process.env.DT_ENV_ID}.live.dynatrace.com/api/v2/otlp/v1/traces`,
    headers: {
      'Authorization': `Api-Token ${process.env.DT_API_TOKEN}`,
    },
  }),
});

sdk.start();
```

5. Entity Mapping

How Dynatrace Maps OTel Data

OTel Resource Attribute	Dynatrace Entity	Notes
`service.name`	Service	Required for service detection
`service.namespace`	Service group	Optional grouping
`host.name`	Host	Links to host entity
`k8s.namespace.name`	K8s namespace	Links to K8s entities
`k8s.pod.name`	Process group	Links to pod

Essential Resource Attributes

Always set these for proper entity mapping:

```
resource = Resource.create({
    # Required
    "service.name": "checkout-api",

    # Recommended
    "service.version": "1.2.3",
    "service.namespace": "ecommerce",
    "deployment.environment.name": "production",

    # For K8s
    "k8s.namespace.name": "checkout",
    "k8s.pod.name": "checkout-api-abc123",
    "k8s.deployment.name": "checkout-api",
})
```

6. Span Attributes for Dynatrace

Dynatrace-Specific Attributes

Attribute	Purpose	Example
`dt.entity.process_group_instance`	Link to PGI	`PROCESS_GROUP_INSTANCE-XXX`
`dt.span.type`	Override span type	`DATABASE`, `MESSAGING`
`dt.source`	Data source	`opentelemetry`

Semantic Conventions Dynatrace Uses

Convention	Dynatrace Feature
`http.*`	HTTP service detection
`db.*`	Database call analysis
`messaging.*`	Message queue tracking
`rpc.*`	RPC call tracking

Example with Rich Attributes

```
with tracer.start_as_current_span("checkout") as span:
    # Standard semantic conventions
    span.set_attribute("http.method", "POST")
    span.set_attribute("http.url", "https://api.example.com/checkout")
    span.set_attribute("http.status_code", 200)
    span.set_attribute("http.route", "/checkout")

    # Business context
    span.set_attribute("order.id", order_id)
    span.set_attribute("order.total", order_total)
```

Complex attribute types (SaaS 1.344)

Forthcoming / rolling out (SaaS 1.344). SaaS 1.344 released 07/27/2026 with a **staged tenant rollout** (from 07/29/2026): OTLP span attributes accept **complex data types** — nested arrays and maps — instead of scalars only.

Verify the release has reached your tenant before you start exporting nested attributes. How a pre-1.344 tenant treats a nested attribute is not something to discover from production trace data; send one span from a test service and confirm the structure arrives intact.

Flat scalar attributes are not superseded. They remain the right default for anything you filter, group, or alert on — they are cheaper to query and they keep cardinality legible (see **OTEL-04 § 8 — Attribute Guidelines**). A nested structure is opaque to a `by:{...}` clause in the way a flat key is not.

Attribute shape	Use when	Example
Flat scalar	The value is a query dimension — you filter, group, sort, or alert on it	`order.tier = "gold"`, `payment.provider = "stripe"`
Nested array / map	The structure *is* the payload — you carry it for context and read it on a single span	A validation-error list, a resolved routing table, a request body echo

Reach for nesting only in the second case. If you find yourself planning to aggregate on something inside a nested attribute, promote that value to its own flat scalar attribute alongside the structure.

7. OTLP Metric Dimensions

Advanced OTLP Metric Dimensions (January 2026)

Dynatrace introduced an opt-in setting for **advanced OTLP metric dimensions**:

Feature	Details
Primary Grail Fields	Metrics can carry Primary Grail Fields as dimensions
Higher Cardinality	Increased cardinality limits for metric dimensions

Feature	Details
Faster Queries	Performance improvements for high-cardinality metrics
Special Characters	Dimension names can include slashes and other special characters

> **Warning:** After enabling advanced OTLP metric dimensions, OTLP metrics will **no longer** be automatically enriched with the `dt.entity.service` dimension. If you use `dt.entity.service` in SLOs, alerts, or dashboards, update your queries to filter using the underlying attributes (e.g., `service.name`) instead.

Scope Attributes as Dimensions

When **Add Meter name and version as metric dimensions** is enabled:
- `otel.scope.name` (instrumentation library/scope name)
- `otel.scope.version` (instrumentation library/scope version)

These are automatically added as dimensions to ingested OTLP metrics.

OTLP Auto-Configuration for Kubernetes

Dynatrace Operator (v1.8+) supports **automatic OTLP exporter configuration** for K8s workloads:

Feature	Details
Auto-injection	Environment variables injected into application pods at startup
Routing	Traffic routes through in-cluster ActiveGate if available
Control	Per-pod opt-in/out via `otlp-exporter-configuration.dynatrace.com/inject` annotation
Docs	OTLP Auto-Config

For more details, see Configure OTLP Metrics Ingestion.

```
// Verify OTel traces are arriving
fetch spans, from:-1h
| filter isNotNull(otel.library.name)
| summarize count = count(), by:{service.name, otel.library.name}
| sort count desc
| limit 20
```

```
// Check recent OTel spans
fetch spans, from: now() - 1h
| filter isNotNull(otel.library.name)
| fields timestamp, service.name, span.name, duration
| sort timestamp desc
| limit 30
```

```
// View OTel data by instrumentation scope
fetch spans, from:-1h
| filter isNotNull(otel.scope.name)
| summarize count = count(), by:{otel.scope.name}
| sort count desc
| limit 10
```

8. Verification

Verify Data in Dynatrace

1. **Services**: Go to **Services** - OTel services appear with OTel icon
2. **Traces**: Go to **Distributed Traces** - Filter by service
3. **Metrics**: Go to **Metrics** - Search for custom metric names
4. **Logs**: Go to **Logs & Events** - Filter by trace_id

Troubleshooting Checklist

Check	How
Token scopes	Settings > Access tokens
Collector logs	`otelcol-contrib --config config.yaml`
Network connectivity	`curl -v https://{env}.live.dynatrace.com/api/v2/otlp`

Check	How
service.name set	Check resource attributes

Common Issues

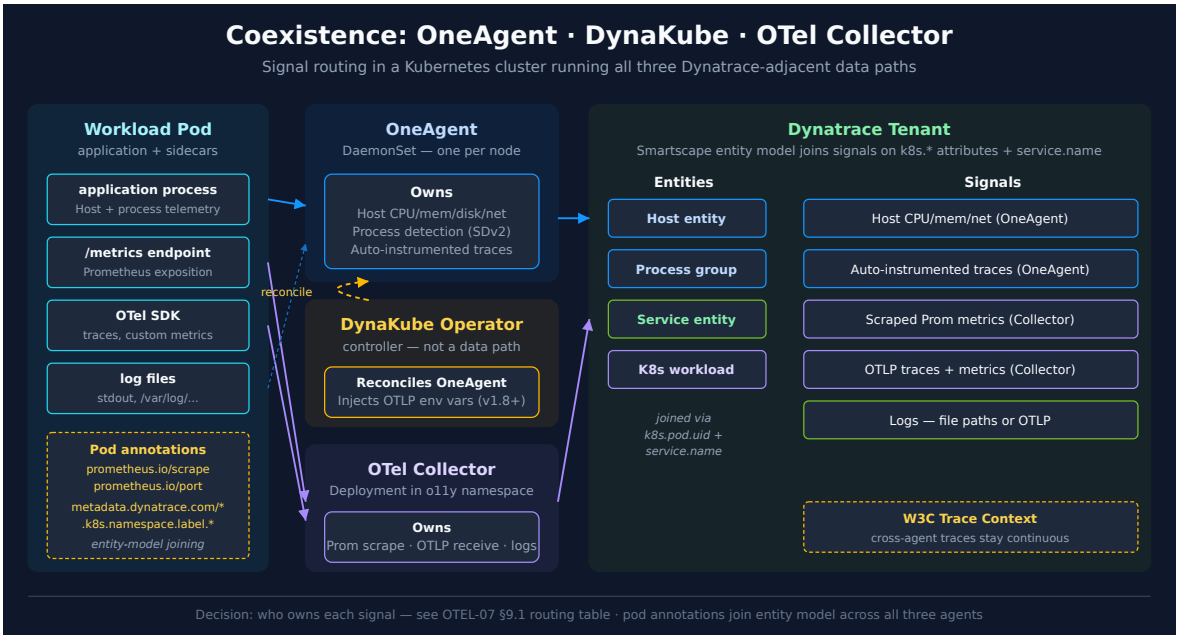
Issue	Cause	Fix
401 Unauthorized	Invalid token	Check token, regenerate
403 Forbidden	Missing scope	Add required scopes
No data	service.name missing	Add resource attribute
Partial data	Network timeout	Check batch size, retry config

9. Coexistence with OneAgent and DynaKube

In Kubernetes you may have three Dynatrace-adjacent data paths active at once:

- **OneAgent** — host-process auto-instrumentation, deployed by the DynaKube operator.
- **OTel Collector** — your own deployment, scraping Prometheus targets or accepting OTLP from apps.
- **OTLP auto-config** — DynaKube Operator (v1.8+) can inject OTLP exporter env vars into application pods.

These do not conflict on the wire, but they overlap on what telemetry they produce. The decision is **which agent handles which signal**.



9.1. Signal-Routing Decision Table

Signal type	Preferred path	Why
Host CPU / memory / disk / network	OneAgent (DynaKube)	Auto-discovers; tightly integrated with the Smartscape host entity
Process-level metrics, process group detection	OneAgent	SDV2 entity detection requires the agent
Application traces (OneAgent-supported language)	OneAgent	Auto-instrumentation, no code change
Application traces (unsupported language or fine-grained custom)	OTel SDK → Collector → OTLP	OTel covers languages OneAgent does not
Prometheus-exposed metrics from third-party / business apps	OTel Collector `prometheus` receiver → OTLP	OneAgent does not scrape Prometheus
Custom application metrics	OTel SDK → Collector → OTLP	Same as Prometheus path
Logs from OneAgent-instrumented hosts	OneAgent	Auto-detection from log paths

Signal type	Preferred path	Why
Logs from OTel-only apps	OTel SDK → Collector → OTLP	OneAgent does not pick them up without the agent installed

9.2. Entity Model Joining via `metadata.dynatrace.com/*`

When the OTel Collector scrapes a pod, Dynatrace does not automatically know how to link the resulting metrics to existing entities (the service detected by OneAgent, the K8s deployment in DynaKube's view). The `k8sattributes` processor plus a pod-annotation convention solves this:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: myapp
spec:
  template:
    metadata:
      annotations:
        # Standard Prometheus scrape contract (see OTEL-05 §6.2)
        prometheus.io/scrape: "true"
        prometheus.io/port: "8000"

        # Dynatrace entity-metadata hints — picked up by k8sattributes
processor
        metadata.dynatrace.com/k8s.namespace.label.domain: sandbox
        metadata.dynatrace.com/k8s.namespace.label.otel.service.name: myapp
```

The collector's `k8sattributes` processor pulls these annotations into the resource attributes of every scraped metric. A downstream `transform` processor can then merge the JSON blob into top-level attributes. Result: scraped metrics carry `service.name = "myapp"` matching what OneAgent or the OTel SDK produced for the same workload — and Dynatrace's entity model joins them.

9.3. Counter Temporality — `cumulativetodelta`

Prometheus emits **cumulative** counters that only ever increase. Dynatrace prefers **delta** temporality — the per-interval change. Without conversion, every counter looks ever-increasing in Dynatrace and `rate()` math is awkward at query time. The collector's `cumulativetodelta` processor performs the conversion in-pipeline:

```
processors:
  cumulativetodelta:
    max_staleness: 3m
```

For full pipeline shape including this processor see **OTEL-05 §6.5 Full Pipeline**.

9.4. Auth Scheme Reminder

OTLP from a Collector uses `Authorization: Api-Token ${TOKEN}` (classic) or `Authorization: Bearer ${TOKEN}` (Platform Token). DynaKube's data-ingest token follows the same rule — match the scheme to the token prefix. See **§2 Authentication Setup → Token Type and Auth Scheme** for the matrix.

9.5. Trace Context Propagation

OneAgent and OTel both support W3C Trace Context, so a request that crosses agent boundaries produces a single trace in Dynatrace:

```
Service A (OneAgent) → Service B (OTel) → Service C (OneAgent)
                ↓                  ↓
            Dynatrace traces show complete path
```

Configure OTel to use W3C propagation explicitly — some SDKs default to other formats:

```
from opentelemetry.propagate import set_global_textmap
from opentelemetry.propagators.composite import CompositePropagator
from opentelemetry.propagators.w3c.traceparent import W3CTraceparentPropagator
from opentelemetry.propagators.w3c.tracestate import W3CTracestatePropagator

set_global_textmap(CompositePropagator([
    W3CTraceparentPropagator(),
    W3CTracestatePropagator()
]))
```

> **Sources:**
>
> - k8sattributes processor (OpenTelemetry Collector Contrib GitHub)
> - Cumulative to Delta processor (OpenTelemetry Collector Contrib GitHub)
> - Configure OTLP Metrics (DT docs)
> - Dynatrace Operator (Dynatrace GitHub)

> ¯ **Derived:** signal-routing table combines OneAgent's coverage capabilities with OTel's gap-filling role; final placement is engagement-specific

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Summary

In this notebook, you learned:

- Dynatrace OTLP endpoints (HTTP only — gRPC requires Collector)
- Dynatrace OTel Collector distribution for production deployments
- API token creation with required scopes; classic `Api-Token` vs Platform `Bearer` auth schemes
- Complete Collector configuration
- Direct SDK export configuration
- Entity mapping and resource attributes
- Span attributes that enhance Dynatrace analysis
- OTLP metric dimensions changes and auto-configuration
- Verification and troubleshooting
- Coexistence patterns across OneAgent, DynaKube, and the OTel Collector — signal routing, entity-model joining via `metadata.dynatrace.com/*`, counter temporality

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References

- Dynatrace OpenTelemetry
- OTLP API Endpoints
- Dynatrace Collector
- Configure OTLP Metrics
- OTLP Auto-Config for K8s
- Ensure Success with OTel

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.